

Chidubem Onwuchuluba

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Education

Georgia State University, Master of Science (MS) in Computer Science Aug 2025 - Dec 2027
• **GPA:** 3.92/4.0 • **Concentration:** Machine Learning and Software Engineering
• **Coursework:** Machine Learning, Computational Intelligence (Neural Networks, Optimization), Secure and Private AI
Covenant University, Bachelor of Science (BS) in Computer Science Aug 2019 - May 2023
• **GPA:** 3.87/4.0, Magna Cum Laude • **Coursework:** Data Structures and Algorithms, Object-Oriented Programming

Technical Skills

Languages: Python, Java, C++, C#, Go, JavaScript, SQL
CS Fundamentals: Object-Oriented Programming (OOP), Data Structures and Algorithms, design patterns, system design
Cloud & Infrastructure: Microsoft Azure, AWS (EC2, S3, CloudWatch), Docker, Kubernetes, Git, Jenkins, GitLab CI, Linux
Databases & Systems: PostgreSQL, MySQL, MongoDB, Redis; indexing, caching, transactions; distributed systems, concurrency
AI & Agents: OpenAI Agents SDK, A2A protocol, Model Context Protocol (MCP), LLM tool use, multi-agent orchestration
Engineering Practices: Unit/integration/load testing, CI/CD, code review, observability, dependency injection, mypy, Pydantic, Agile

Experience

Software Engineering Intern | Paystack (YC W16, Interswitch Group) Jun 2024 - Dec 2024
• Partnered with product managers and cross-functional teams to define requirements, then designed, developed, and tested RESTful APIs in C#, Go, and Python serving 2M+ daily payment transactions at 99.9% availability; added a Redis caching layer with explicit TTL/invalidation over PostgreSQL, cutting response time 45% with correct payment-state reads
• Owned containerized microservices end-to-end on Docker and Kubernetes across hybrid AWS and Microsoft Azure infrastructure for 50,000+ merchants, from design review to production rollout; as on-call engineer, improved reliability and observability with CloudWatch dashboards and alerting
• Raised deployment frequency 60% with zero-downtime blue-green deploys on Jenkins and GitLab CI; reviewed code and applied best practices across a C#/Go/Python codebase
Software Engineering Intern | Rotawn Energy Jan 2025 - Aug 2025
• Gathered requirements from operations stakeholders and shipped a Visitor Management web app with QR-based authentication and SMS notifications, reducing manual processing time 40%
• Replaced manual reporting with JavaScript data-visualization dashboards, saving 15+ staff hours per week; maintained 99.5% uptime through proactive monitoring and automated backups
Graduate Teaching Assistant | Georgia State University Aug 2025 - Present
• Built Python automation scripts for grading and lab administration serving 110+ students in Computer Organization Programming
• Mentored students through office hours and 1-on-1 debugging sessions; configured Linux/Windows lab environments and resolved system-level issues

Projects

Multi-Agent Trip Planning Network (A2A + MCP) GitHub
Python 3.11+ (async), OpenAI Agents SDK, A2A protocol, MCP, Starlette/uvicorn, httpx, Pydantic
• Built a network of LLM agents that delegate to each other over A2A as English-in/English-out function calls, where one agent's tool call runs another agent's reasoning loop remotely; debugged A2A/MCP interoperability issues
• Applied object-oriented design to split each agent into four reusable layers (domain tools, MCP server, SDK agent, A2A service), so one implementation runs as a unit test, an MCP tool, or a networked service; tiered models to cut inference cost without hurting plan quality
• Engineered for reliability and fault tolerance with fail-fast validation, layered timeouts, and isolated connection pools that prevent head-of-line blocking; 287-test dependency-injected suite (mypy + Pydantic strict) runs in about 10s with no live API calls
PostgreSQL-Backed Real Estate Data Platform Website
Django, PostgreSQL, AWS (EC2, S3, CloudFront), Celery, Redis
• Built and operate a production platform on a PostgreSQL data model with a scheduled Celery sync pipeline on EC2 that keeps listings, images, and metadata consistent via idempotent upserts
• Translate a non-technical client's requirements into shipped features; own schema design, deployment (Nginx/Gunicorn, S3/CloudFront), monitoring, and incident response

Certifications

GitHub Certified: Agentic AI Developer (GitHub/Microsoft) • AWS Certified Solutions Architect • IBM AI Developer